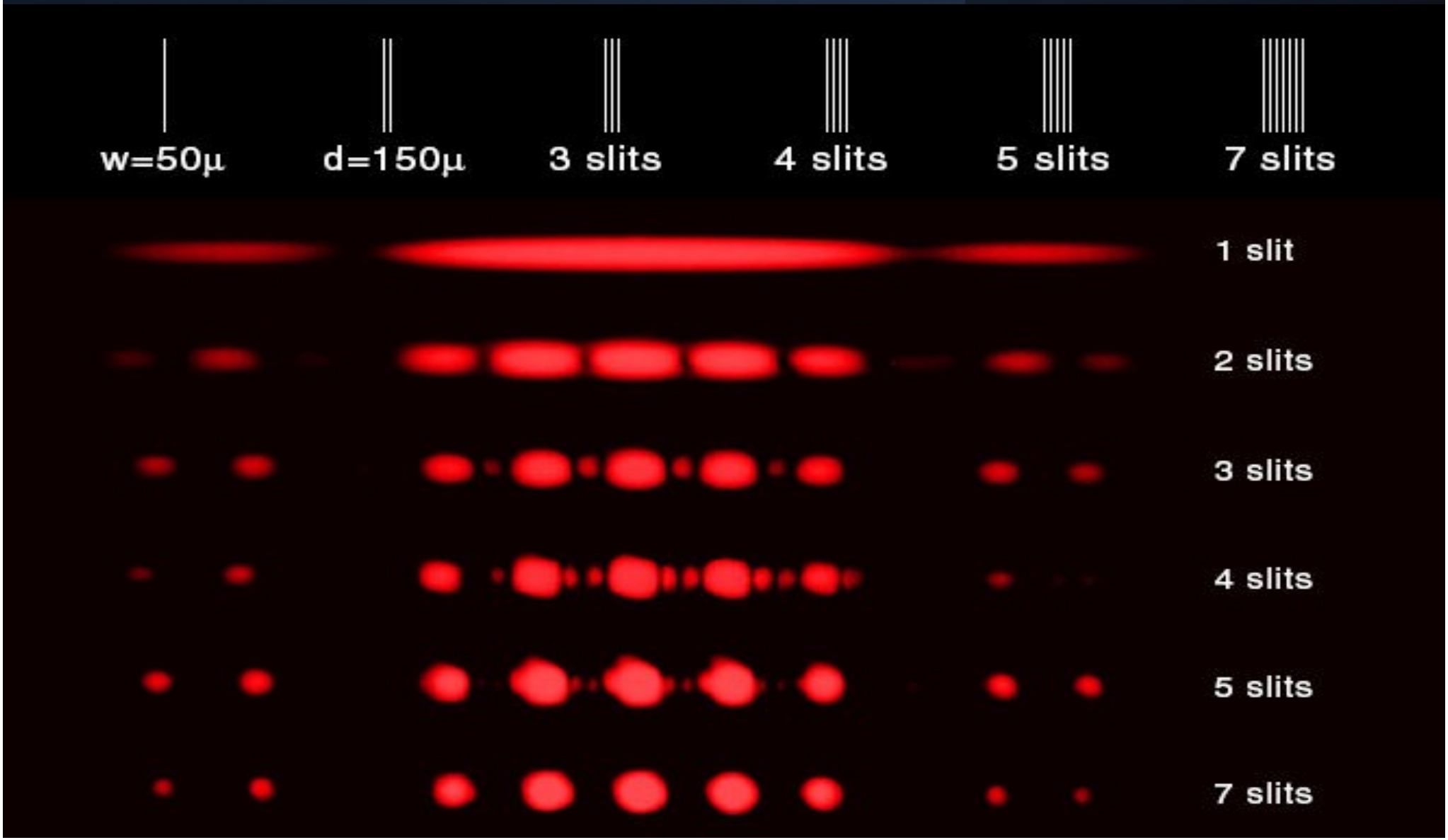
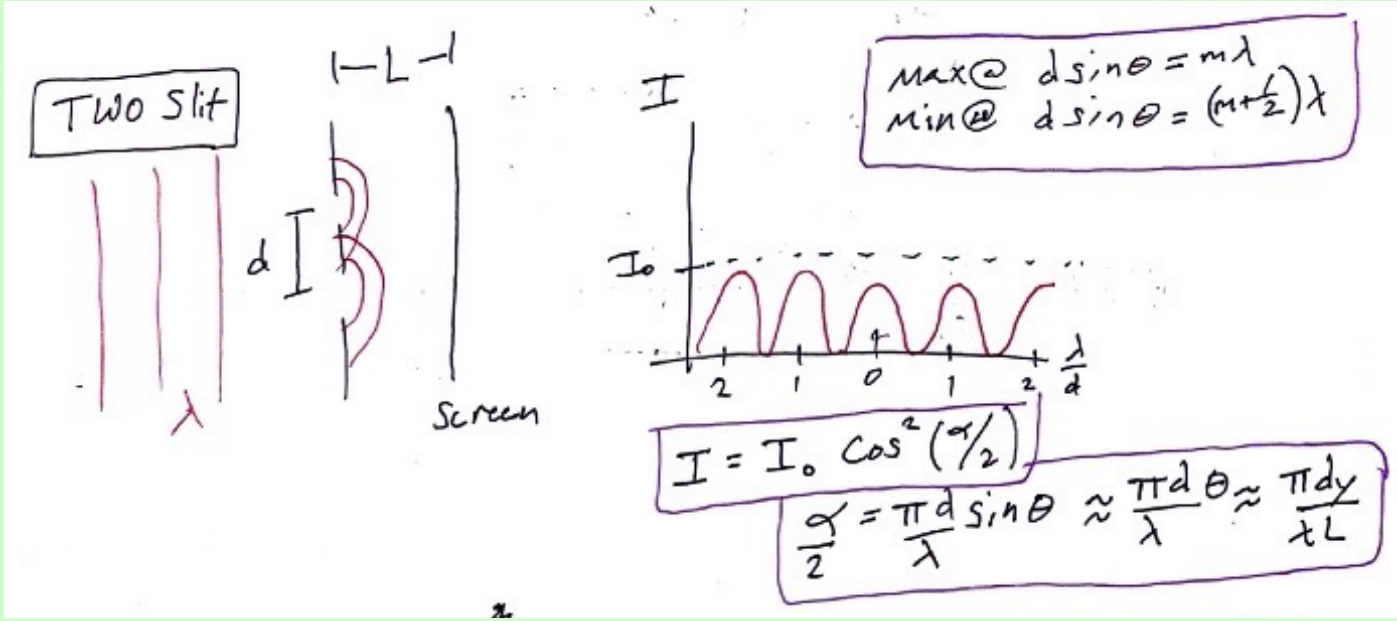
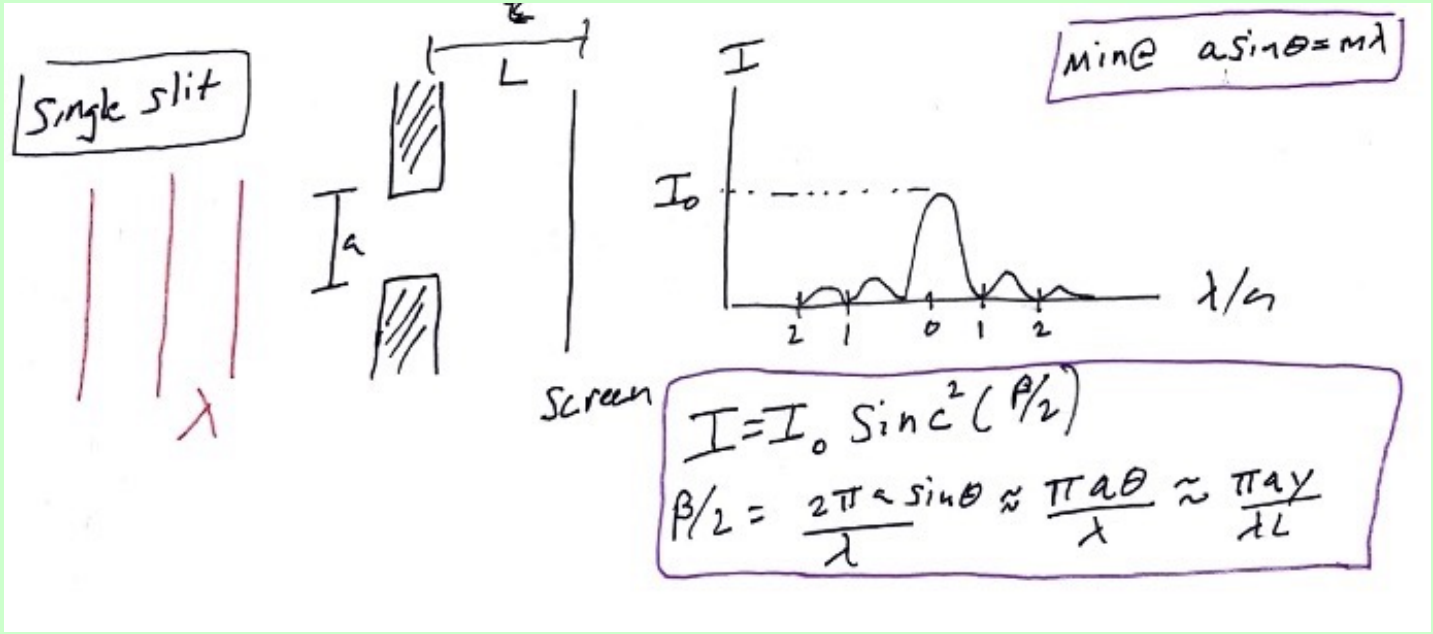
Applications of Diffraction



Review:


Two slit
L
d
λ
Screen
I
I_0
2 1 0 1 2
λ/d
max@ $d\sin\theta = m\lambda$
min@ $d\sin\theta = (m+\frac{1}{2})\lambda$
$I = I_0 \cos^2(\alpha/2)$
$\frac{\alpha}{2} = \frac{\pi d}{\lambda}\sin\theta \approx \frac{\pi d}{\lambda}\theta \approx \frac{\pi d y}{\lambda L}$


Single slit
L
a
λ
Screen
I
I_0
2 1 0 1 2
λ/a
min@ $a\sin\theta = m\lambda$
$I = I_0 \,\mathrm{Sinc}^2(\beta/2)$
$\beta/2 = \frac{2\pi a}{\lambda}\sin\theta \approx \frac{\pi a\theta}{\lambda} \approx \frac{\pi a y}{\lambda L}$

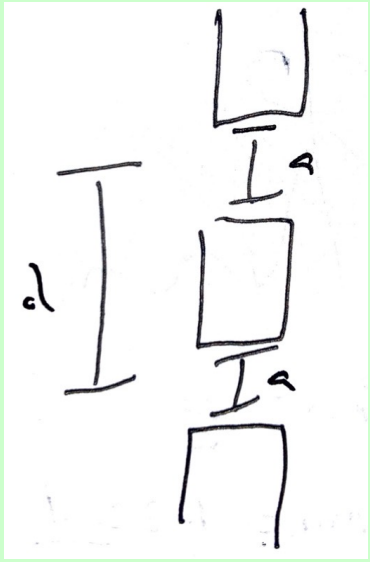
Ex. Light of wavelength 690 nm travels through a slit of width $a = 5.0 \times 10^{-5}$ m. The screen is 0.5 m from the slit.

A. At which angle is the second bright spot, aka the "first secondary maximum"?

B. What is the position of this maximum on the screen?

C. Sketch the single slit diffraction intensity pattern, the write out the intensity pattern as an equation.

It is also possible to have **Two Slit Diffraction**



The slits are each width **a**, but are separated by **d**



The overall envelope now set by diffraction, But inside each diffraction hum are a number of interference fringes.

Diffraction **minima** at $a \sin\theta = m\lambda$

Diffraction **maxima** at $d \sin\theta = m\lambda$

Two Slit Diffraction Intensity $I = I_0 \cos^2\left(\frac{\alpha}{2}\right) \mathrm{Sinc}^2\left(\frac{\beta}{2}\right)$

Ex. What happens if a is increased but everything else is the same?

What happens if d is decreased by everything else is the same?

Ex. Two slits of slit width $a = 1 \times 10^{-4}$ m are separated by $d = 2 \times 10^{-4}$ m.

A. Find the first two diffraction minima and draw the diffraction envelope

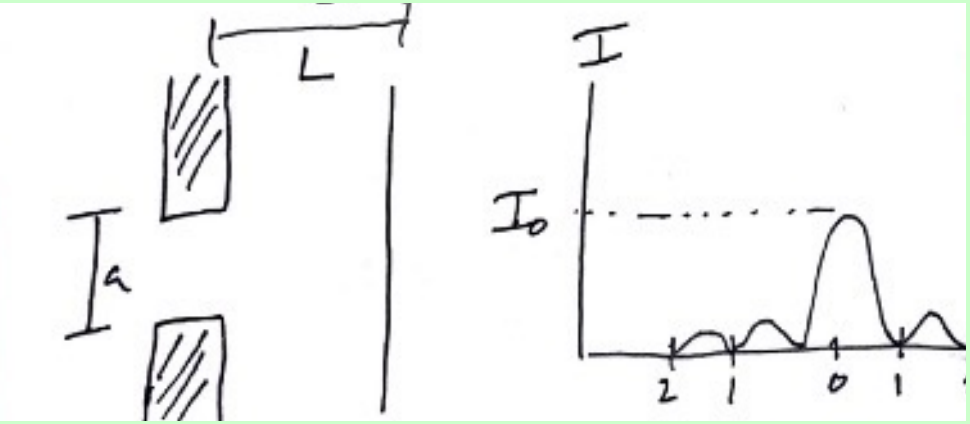
B. Find the first two Interference maxima, and draw the interference fringes

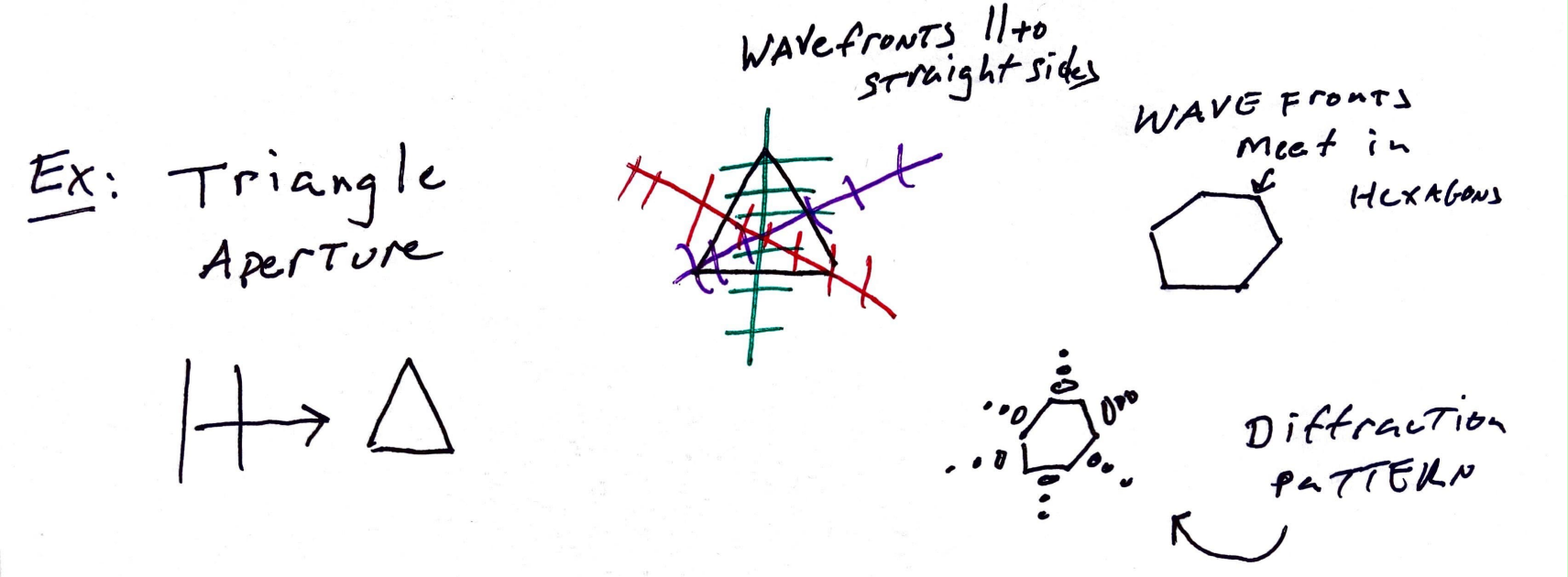
C. If $I_0 = 3\ W/m^2$, what is the equation for the Intensity pattern?

2D Apertures

We've talked about diffraction pattern through single slits, but what happens when the plane wave shines through a 2D aperture?

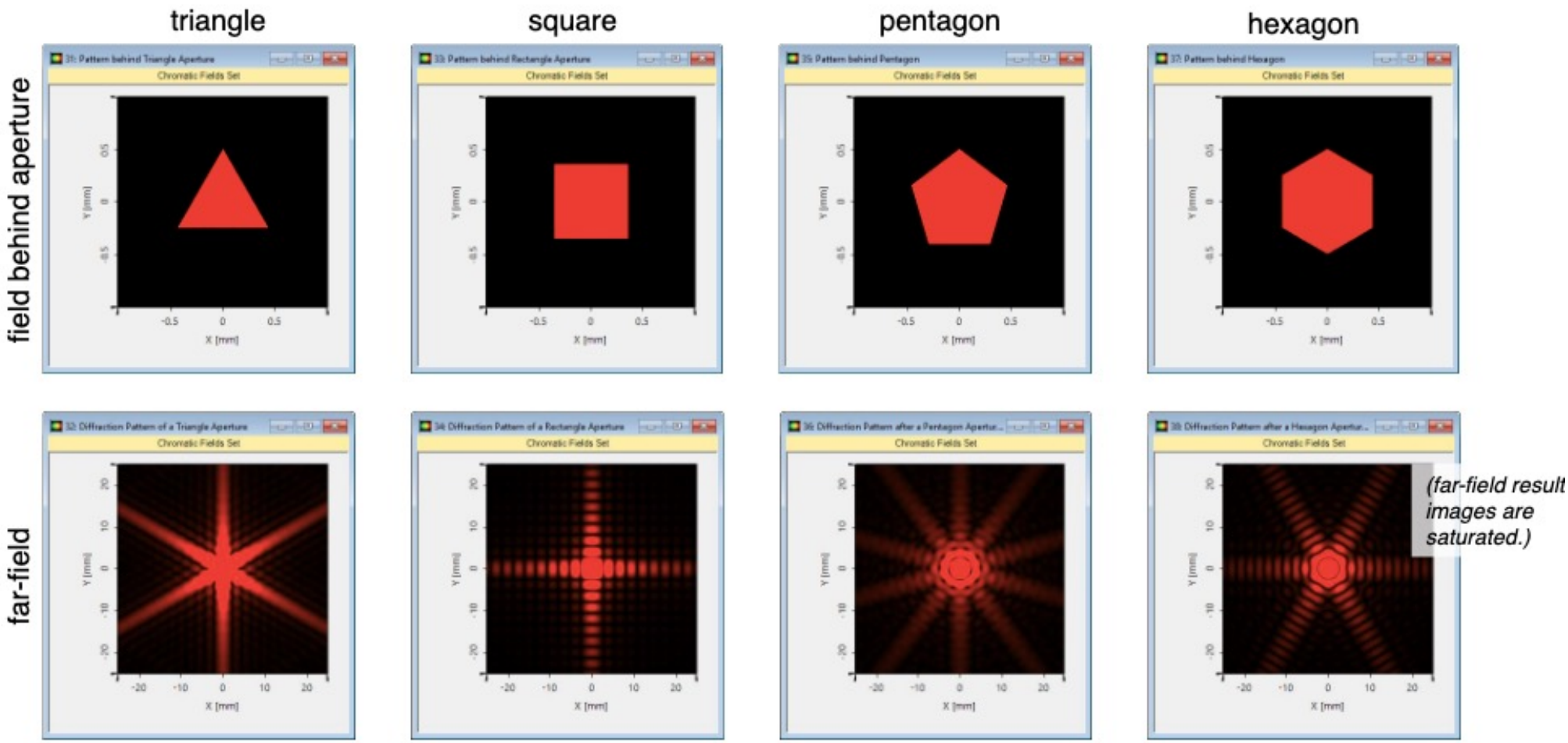
1 D aperture



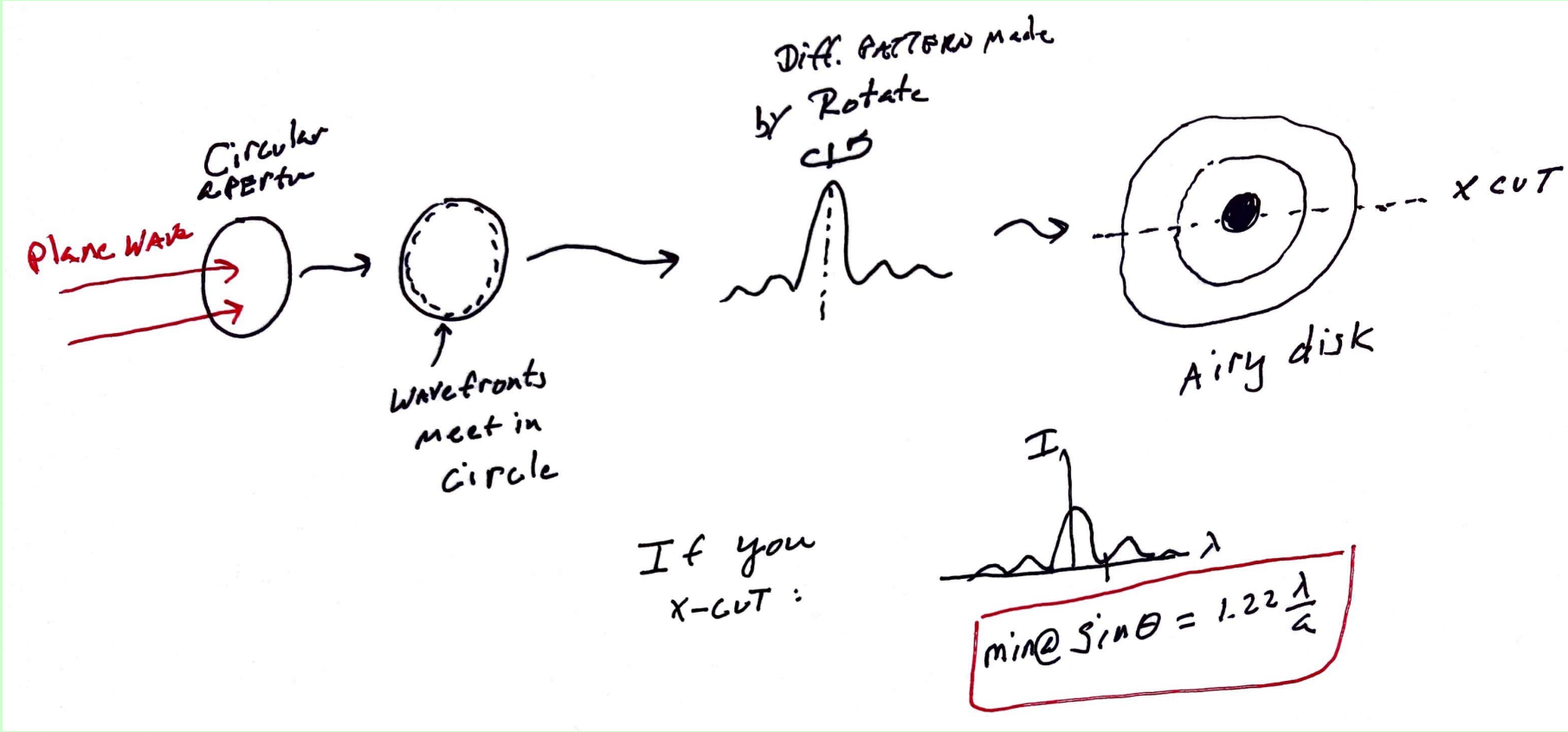


Ex. Predict Diffraction pattern of a rectangle, hexagon, pentagon, circle

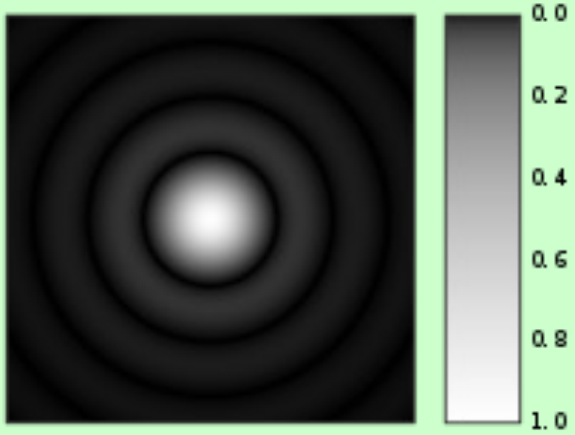
Diffraction patterns of polygons



Circular Apertures make Diffraction Patterns called **Airy Disks**



Computer generated

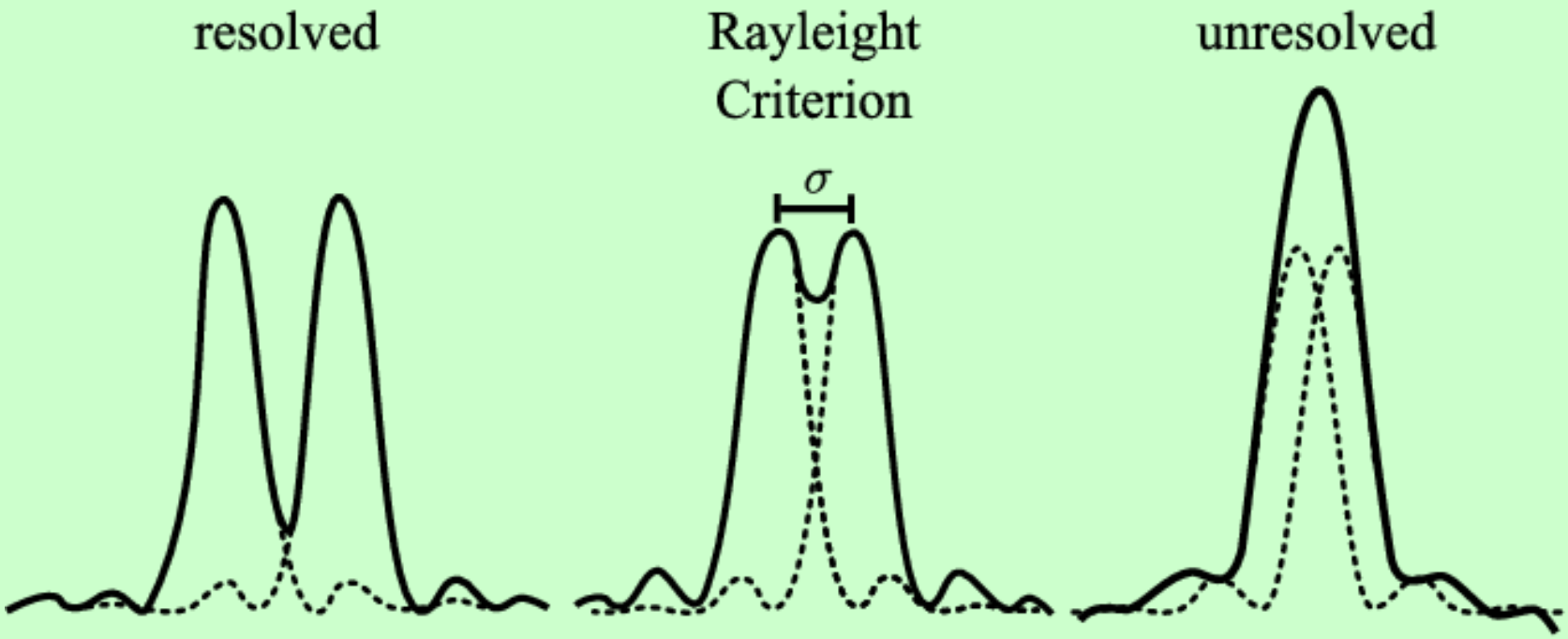


$$\sin\theta = 1.22\left(\frac{\lambda}{a}\right)$$

Ex. Keck is a telescope with primary mirror that is a = 10 m in diameter. Find the angular size of the central maximum in its airy disk

The Rayliegh Criterion

If two stars are super close to eachother, how
Can you determine if you can resolve them?



The Rayleigh criterion says that two stars are resolveable if the peak of one airy disk
Is overlapping with the first minimum of the other. Thus,

$$\Delta\theta = 1.22\left(\frac{\lambda}{a}\right)$$

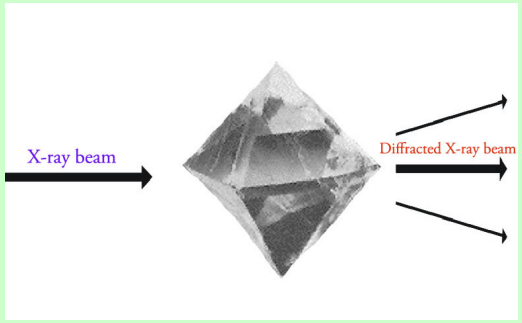
Ex. Cosmic Hoax?!?

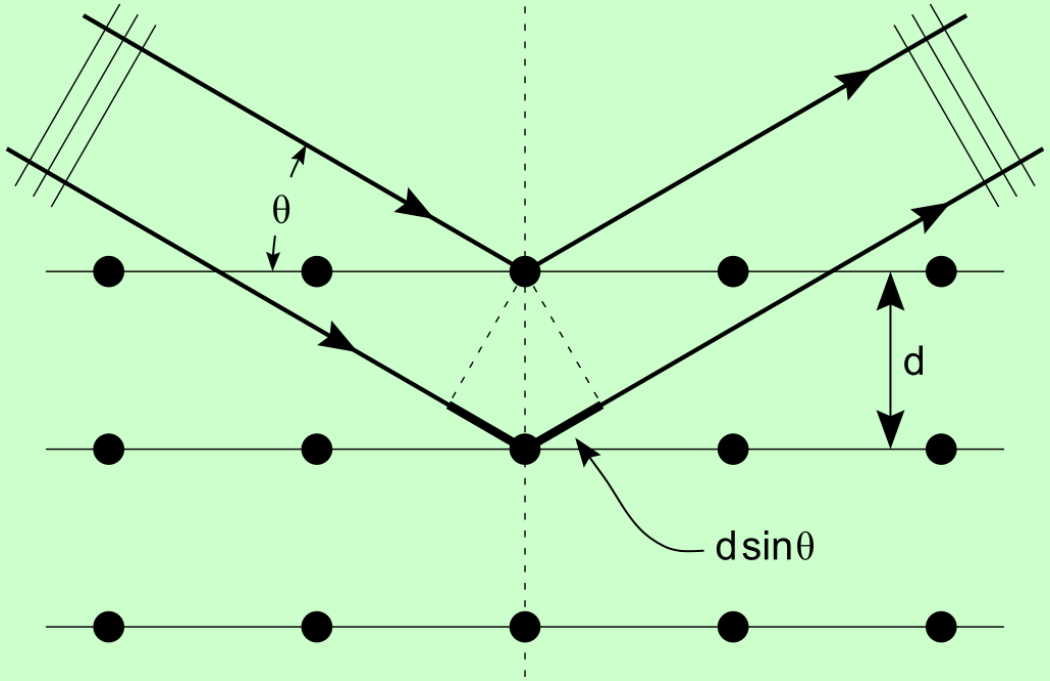
Maybe you've heard we can't even see the flags the astronauts planted on the moon. Is this proof of man never having gone there?

A. The moon is d = 3.84×10^8 m away. Suppose the flag is 1m x 1m cross section
What angle does the flag subtend as seen from the Earth?

B. What is the minimum angle the Rayliegh criterion says should be resolvable by Hubble?

X-ray Diffraction: Spacing of atoms in crystal lattice similar to x-ray wavelengths





Suppose I shoot in x-rays at some angle, which angle will they come out?

Maxima at: $\Delta r = m\lambda$

$2d\sin\theta = m\lambda$

Bragg Equation:

$$\boxed{m\lambda = 2d\sin\theta}$$

Ex. If x-rays of wavelength 1.2 Angstroms are shot at a crystalline lattice with an inter-atomic spacing of d = 0.8 angstroms, what angles will see the first bright maximum of diffracted x-rays?

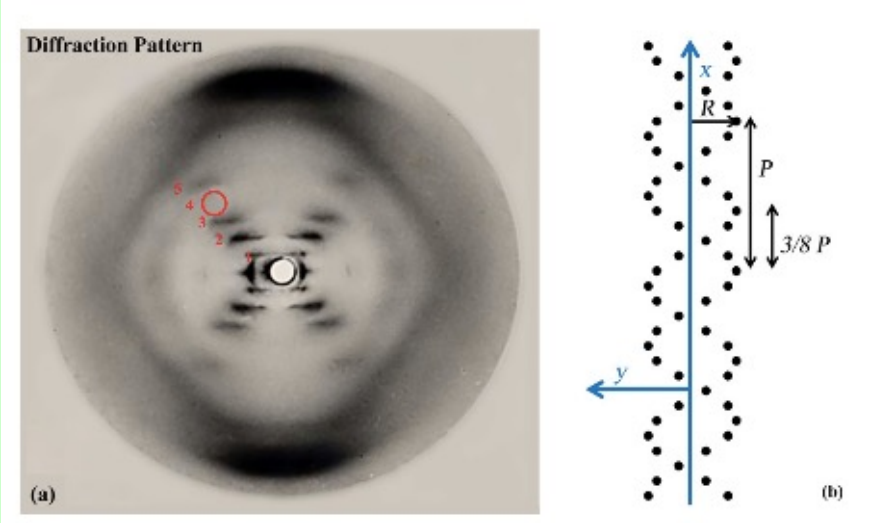
Rosalind Franklin and the Helical Structure of DNA

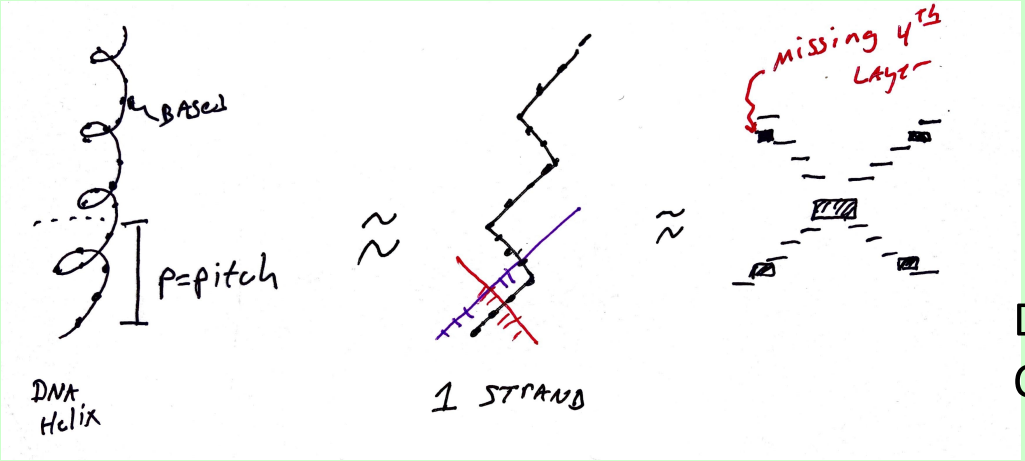


1920-1958

Used x-ray crystallography to get a diffraction pattern of DNA

Photo 51





P = Pitch. Strand Wavelength
Distance btwn bases h = 1/10 P
Distance btwn strands x = 3/8 P

Destructive Int from second strand
Causes missing fourth

Photo 51

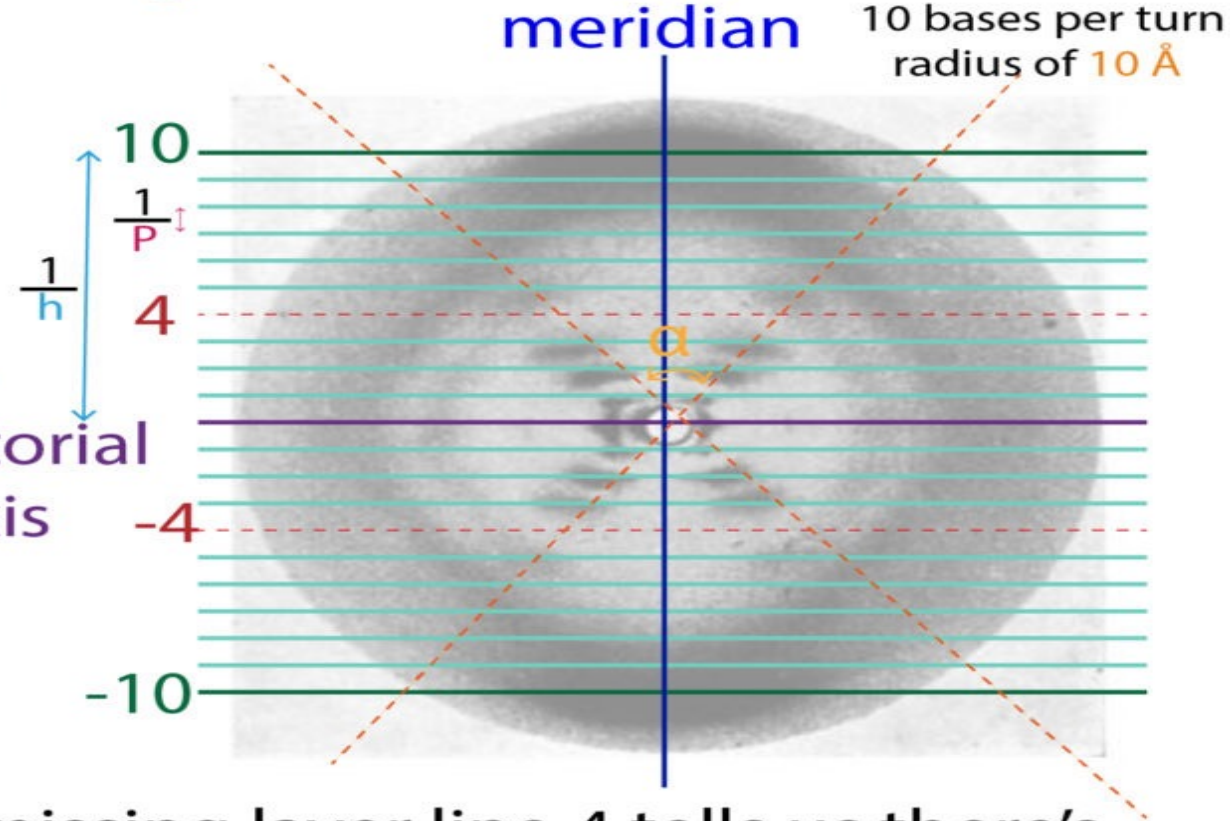
fiber diffraction pattern of B-form DNA



strong meridional arcs (smears at top and bottom) come from the horizontal bases. The distance between them and the center tells us about the interbase distance, h

spacing betwen layer lines tells us about the helical pitch, P

angle tells us about the radius



missing layer line 4 tells us there's a second strand causing destructive interference